

UCLA Math 131A, Winter 2025
Midterm 1 Practice

Instructor: Linfeng Li

Monday, January 29

Please print your name and student ID below.

Name: _____

Student ID: _____

Please read all of the following rules before proceeding.

- The exam worth 40 points in total.
- Show all your work, and precisely state any theorems that you use.
- You are not allowed to use calculators or notes during the test, but you may use one-page, double-sided, and handwritten cheat sheet.
- Good luck!

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
Total:	40	

1. (10 points) Circle TRUE or FALSE for each of the following statement. No justification is needed.

- (i) ☒ True False The negation of the statement “ $P \implies Q$ ” is “ P and $\neg Q$ ”.
- (ii) True ☒ False The set $\{a \in \mathbb{Q} : a < 1\}$ attains both maximum and supremum.
- (iii) ☒ True False If $S \subseteq \mathbb{R}$ is a set with finitely many elements, then S has both maximum and minimum.
- (iv) ☒ True False Let $S \subseteq \mathbb{R}$ be nonempty. If the maximum of S exists, then the maximum is also the supremum of S .
- (v) True ☒ False The negation of the statement “ $\exists r \in \mathbb{R}$, s.t. $\forall x \in [a, b], f(x) < r$ ” is “ $\forall r \in \mathbb{R}, \exists x \in [a, b]$, s.t. $f(x) < r$ ”

2. (10 points) Use induction to prove the sum-of-squares formula

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Solution: Let P_n be the statement

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

The base case P_1 is “ $1^2 = \frac{1 \times 2 \times 3}{6}$ ” which clearly holds. Suppose that P_n holds for some $n \in \mathbb{N}$. We aim to prove P_{n+1} holds. We compute

$$\begin{aligned} 1^2 + 2^2 + \cdots + n^2 + (n+1)^2 &= \sum_{k=1}^n k^2 + (n+1)^2 \\ &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \left(\frac{n(2n+1)}{6} + (n+1) \right) (n+1) \\ &= \frac{2n^2 + 7n + 6}{6} \cdot (n+1) = \frac{(n+1)(n+2)(2(n+1)+1)}{6}. \end{aligned}$$

Therefore, P_{n+1} holds. By induction, P_n holds for any $n \in \mathbb{N}$.

3. (a) (5 points) Let r be a nonzero rational number and x be an irrational number. Prove that rx is irrational.
- (b) (5 points) Let $a, b \in \mathbb{R}$. Prove that if $a \leq b + \epsilon$ for any $\epsilon > 0$, then $a \leq b$.

Solution: (a) For the sake of contradiction, we assume that rx is rational. By definition,

$$rx = \frac{m}{n}, \text{ where } m, n \in \mathbb{Z} \text{ and } n \neq 0.$$

Since r is a rational number, we have

$$r = \frac{p}{q}, \text{ where } p, q \in \mathbb{Z} \text{ and } q \neq 0.$$

It follows that

$$x = \frac{mq}{np}.$$

It is clear that mq and np are integers and the denominator is nonzero. Hence, x is a rational number. This is a contradiction.

(b) For the sake of contradiction, we assume that $a > b$. Let $\epsilon = (a - b)/2 > 0$. By assumption, we have

$$a \leq b + \frac{a - b}{2} = \frac{a + b}{2} \implies 2a \leq a + b \implies a \leq b.$$

This contradicts with $a > b$.

4. (a) (5 points) Let E, F be two nonempty bounded subsets of $\mathbb{R}$. Prove that

$$\sup(E \cup F) \geq \max\{\sup(E), \sup(F)\}.$$

- (b) (5 points) Prove that the greatest lower bound of the set $\{\frac{1}{n} : n \in \mathbb{N}\}$ is 0.

Solution:

- (a) $\forall s \in E \cup F$, we have $s \leq \sup(E \cup F)$ by definition. In particular, $\sup(E \cup F) \geq s, \forall s \in E$. Thus $\sup(E \cup F)$ is an upper bound of E . Since $\sup(E)$ is the least upper bound of E , we deduce that $\sup(E \cup F) \geq \sup(E)$. Similarly, it follows that $\sup(E \cup F) \geq \sup(F)$. Therefore, we conclude that $\sup(E \cup F) \geq \max\{\sup(E), \sup(F)\}$.
- (b) Let $S = \{\frac{1}{n} : n \in \mathbb{N}\}$. We infer that 0 is a lower bound of S since $0 \leq \frac{1}{n}, \forall n \in \mathbb{N}$. By the completeness axiom, the set S attains a greatest lower bound $m = \inf S$. It is clear that $m \geq 0$ since 0 is a lower bound. For the sake of contradiction, we assume that $m > 0$. There exists a sufficiently large integer $N > 1/m$, from where we get $1/N < m$. The element $1/N$ belongs to the set S , which contradicts with the fact that m is a lower bound of S . Therefore, the assumption $m > 0$ fails and we get $m \leq 0$. Since $m \geq 0$ and $m \leq 0$, we conclude that $m = 0$.